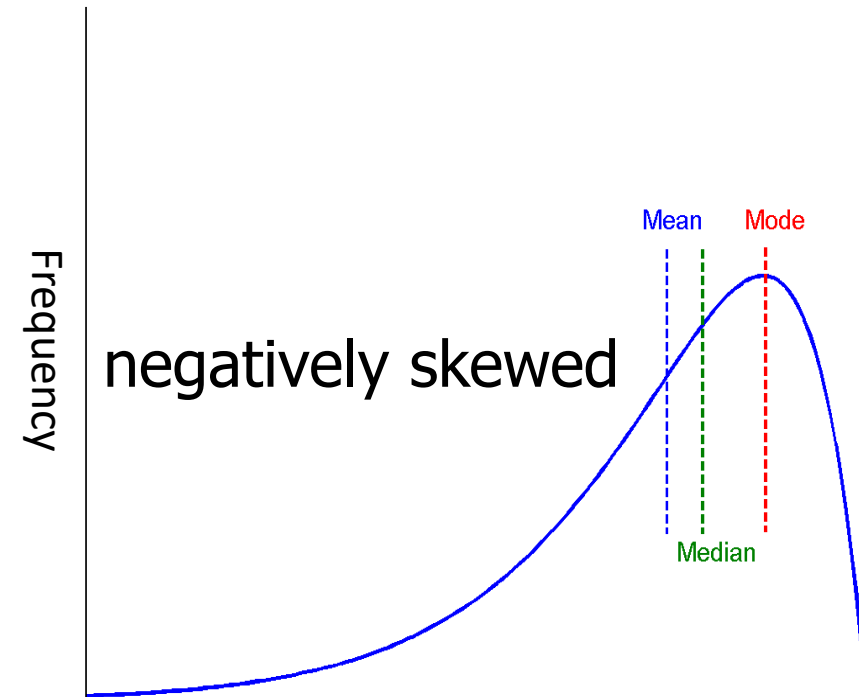
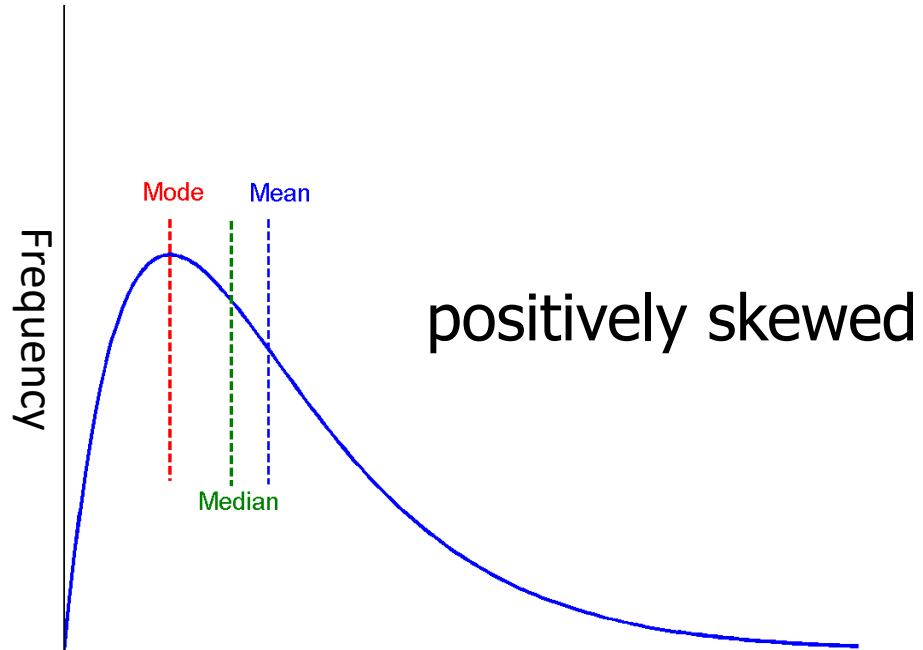
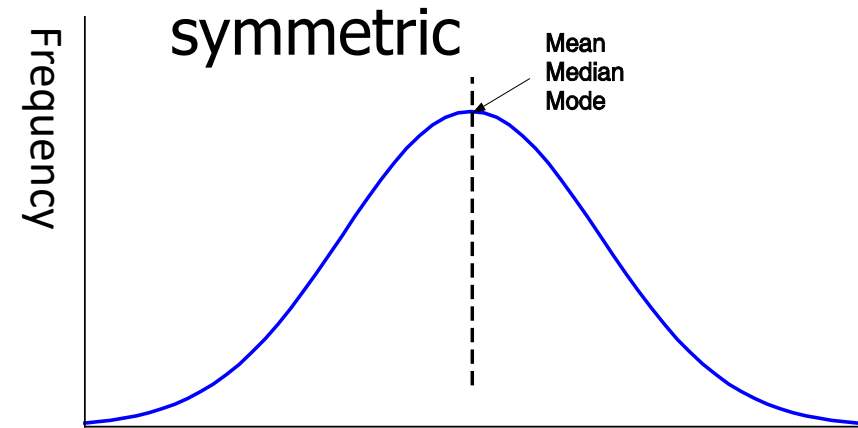


Symmetric vs. Skewed Data

- Median, mean and mode of symmetric, positively and negatively skewed data



Properties of Normal Distribution Curve

← — — — — Represent data dispersion, spread — — — — →

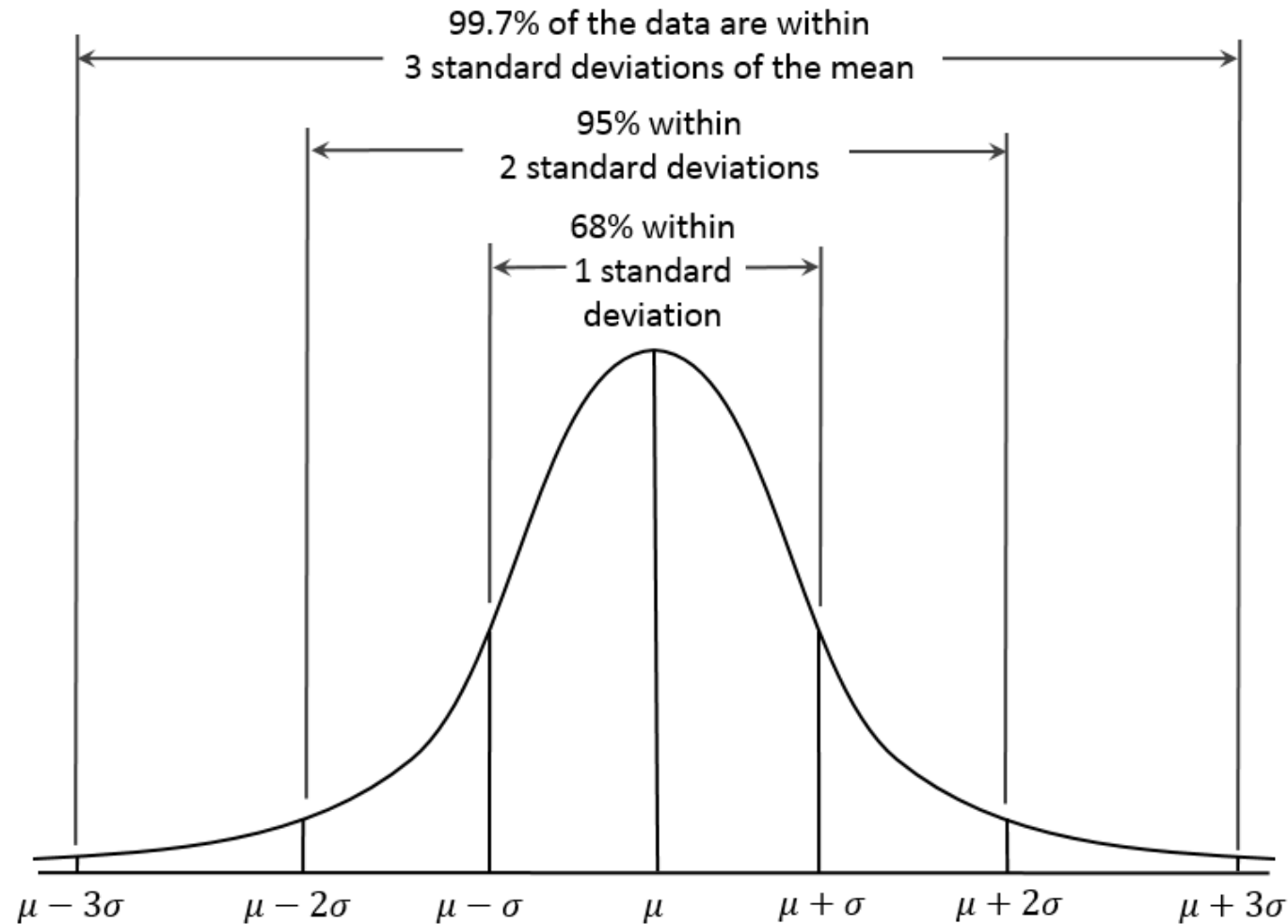


Image Source:

https://upload.wikimedia.org/wikipedia/commons/a/a9/Empirical_Rule.PNG

Represent central tendency

Measures Data Distribution

□ Variance and standard deviation (*sample: s , population: σ*)

□ **Variance:** (algebraic, scalable computation)

□ Q: Can you compute it incrementally and efficiently?

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - \frac{1}{n} \left(\sum_{i=1}^n x_i \right)^2 \right]$$

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^n (x_i - \mu)^2 = \frac{1}{N} \sum_{i=1}^n x_i^2 - \mu^2$$

Note: The subtle difference of formulae for sample vs. population

- n : the size of the sample
- N : the size of the population

□ **Standard deviation s (or σ)** is the square root of variance s^2 (or σ^2)